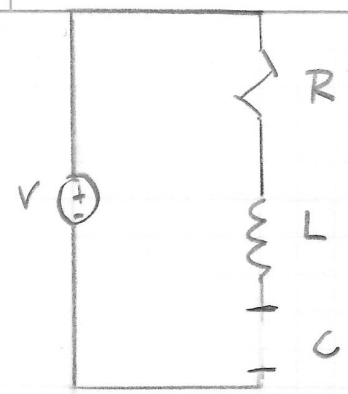
I RLC Circuit



$R = 1 \quad L = 2 \quad C = 3$

$$\dot{x} = \begin{bmatrix} -1/2 & -1/2 \\ 1/3 & 0 \end{bmatrix} x + \begin{bmatrix} 1/2 \\ 0 \end{bmatrix} u$$

0: $x_d = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ ← chosen desired state

1) $z = x - x_d$

2) $B u_{ff} + A x_d - \dot{x}_d = 0$

$$\begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix} u_{ff} + \begin{bmatrix} -1/2 & -1/2 \\ 1/3 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Top Row $\Rightarrow \frac{1}{2} u_{ff} + (-1/2 - 1) - 0 = 0 \Rightarrow \frac{1}{2} u_{ff} = \frac{3}{2} \Rightarrow \underline{\underline{u_{ff} = 3}}$

Bottom Row $\Rightarrow 0 u_{ff} + 1/3 - 0 = 0 \Rightarrow 1/3 = 0$ <u>Problematic</u>

Take away: Not always possible to converge to desired value

$\Rightarrow$ There is no feed-forward term to make x_d an equilibrium point

II RC Circuit w/ different x_d

0: $\dot{x} = \begin{bmatrix} -1/2 & -1/2 \\ 1/3 & 0 \end{bmatrix} x + \begin{bmatrix} 1/2 \\ 0 \end{bmatrix} u \qquad x_d = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

1: $z = x - x_d$

2: $B u_{ff} + A x_d - \dot{x}_d = 0$

$$\Rightarrow \begin{bmatrix} -1/2 & -1/2 \\ 1/3 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 1/2 \\ 0 \end{bmatrix} u_{ff} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1/2\, u_{ff} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \underline{\underline{u_{ff} = 2}}$$

3: $u = \hat{u} + u_{ff} \qquad \dot{z} = \dot{x} - \dot{x}_d$

$$\dot{z} = \dot{x} - \dot{x}_d = Ax + Bu + \cancelto{0}{\dot{x}_d} = A(z + x_d) + Bu$$

$$\Rightarrow \dot{z} = Az + B\hat{u} + B u_{ff} + A x_d$$

$$= Az + B\hat{u} + \begin{bmatrix} 1/2 \\ 0 \end{bmatrix} 2 + \begin{bmatrix} -1/2 & -1/2 \\ 1/3 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$= Az + B\hat{u} + \begin{bmatrix} 1 - 1 \\ 0 + 0 \end{bmatrix} = \underline{\underline{Az + B\hat{u}}}$$

We canceled out the terms we did not want

4: $M = [B \;\; AB]$

$\text{rank}(M) = 2 \Rightarrow$ <u>Fully Controllable</u>

5: $z_{1max} = 2;\ z_{2max} = 4;\ \hat{u}_{max} = 0.5$

$$Q = \begin{bmatrix} 1/z_{1max}^2 & 0 \\ 0 & 1/z_{2max}^2 \end{bmatrix} \qquad R = 1/u_{max}^2$$

$\hat{u} = -kz$ where $k = lqr(A, B, Q, R)$

6: $u = u_{ss} - Kz = \boxed{2 - K(x - x_d)}$

III Smooth Differential Drive

$$\dot{x} = \begin{bmatrix} r/2\,(u_r + u_l)\cos\theta \\ r/2\,(u_r + u_l)\sin\theta \\ r/L\,(u_r - u_l) \\ u_r \\ u_l \end{bmatrix} \qquad x(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad x_d = \begin{bmatrix} 3 \\ 4 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

0: Let's linearize about x_d

$$x = \begin{bmatrix} q_1 \\ q_2 \\ \theta \\ \omega_r \\ \omega_l \end{bmatrix}$$

$$V = \frac{r}{2}(u_r + u_l)$$

$$\omega = r/L\,(u_r - u_l)$$

$$\frac{\delta f}{\delta x} = \begin{bmatrix} 0 & 0 & -V s\theta & r/2\, c\theta & r/2\, c\theta \\ 0 & 0 & V s\theta & r/2\, s\theta & r/2\, s\theta \\ 0 & 0 & 0 & r/L & -r/L \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{where } x^{eq} = x_d, \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Used small angle assumption for $s\theta$ & $c\theta$ for $\theta = 0$

$$\Rightarrow A = \frac{\delta f}{\delta x}\Big|_{\substack{u^{eq} = 0 \\ x^{eq} = x^d}} = \begin{bmatrix} 0 & 0 & 0 & r/2 & r/2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & r/L & -r/L \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B = \frac{\delta f}{\delta u}(x^{eq}, u^{eq}) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow \delta\dot{x} = A\delta x + B\delta u \qquad x^{eq} = x^d \quad u^{eq} = 0$

4: $\Gamma = [B \;\; AB \;\; A^2B \;\; A^3B \;\; A^4B]$; $\text{rank}(\Gamma) < 5$ ← Not completely controllable

HW: What is controllable in this system?

↳ find a controllable decomposition

(Probably ω_r ω_l are controllable states and the rest are not)

IV Smooth Differential Drive - Velocity Control

$$\dot{x} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \\ \omega \\ u_r \\ u_l \end{bmatrix} \qquad V_d = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}$$

0: $x = \begin{bmatrix} v \\ \omega \end{bmatrix} \qquad \dot{x} = \begin{bmatrix} r/2\,(u_r + u_l) \\ r/L\,(u_r - u_l) \end{bmatrix} = [0]x + \begin{bmatrix} r/2 & r/2 \\ r/L & -r/L \end{bmatrix} u$

1: $z = x - x_d$

2: $Bu_{ff} + A\cancel{x_d}^{\,0} - \cancel{\dot{x}_d}^{\,0} = 0 \quad \Rightarrow Bu_{ff} = 0$

$\Rightarrow \underline{\underline{u_{ff} = 0}}$

3: $u = \hat{u} + \cancel{u_{ff}}^{\,0} = \hat{u} \qquad \dot{z} = B\hat{u} \quad$ let $\hat{u} = -k$

4: $\Gamma = [B \;\; AB] \qquad \text{rank}(\Gamma) = 2$ Complete Controllability

5: $\lambda = -1, -2 \qquad k = \text{place}\left(A, B, \begin{bmatrix} -1 \\ -2 \end{bmatrix}\right)$

6: $u = \cancel{u_{ff}}^{\,0} - kz = -kz = -k(x - x_d) = -k\left(\begin{bmatrix} v \\ \omega \end{bmatrix} - \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}\right)$

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V) Point Control: Reference Trajectory

Suppose we can directly control the accelerations of a point:

$$x = \begin{bmatrix} q_1 \\ q_2 \\ \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} \qquad q_d \in \mathbb{R}^2 \quad \dot{q}_d = \mathbb{R}^2 \quad \ddot{q}_d \in \mathbb{R}^2$$

and are known values

$$\dot{x} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} u = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ I \end{bmatrix} u$$

0: $x_d = \begin{bmatrix} q_d \\ \dot{q}_d \end{bmatrix} \qquad \dot{x}_d = \begin{bmatrix} \dot{q}_d \\ \ddot{q}_d \end{bmatrix}$

1: $z = x - x_d$

2: $B u_{ff} + A x_d - \dot{x}_d = \begin{bmatrix} 0 \\ I \end{bmatrix} u_{ff} + \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix} \begin{bmatrix} q_d \\ \dot{q}_d \end{bmatrix} - \begin{bmatrix} \dot{q}_d \\ \ddot{q}_d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 0 \\ u_{ff} \end{bmatrix} + \begin{bmatrix} \cancel{\dot{q}_d} \\ 0 \end{bmatrix} - \begin{bmatrix} \cancel{\dot{q}_d} \\ \ddot{q}_d \end{bmatrix} = \begin{bmatrix} 0_{2\times1} \\ 0_{2\times1} \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ u_{ff} - \ddot{q}_d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \underline{\underline{u_{ff} = \ddot{q}_d}}$$

3: $u = \hat{u} + u_{ff} \rightarrow \dot{z} = Az + B\hat{u}$

4: $\Gamma = [B \;\; AB \;\; A^2B \;\; A^3B]$ and $\text{rank}(\Gamma) = 4$ Fully Controllable

5: Let $\lambda = -1, -2, -3, -4$

$\hat{u} = -kz = -k(x - x_d)$; $k = \text{place}(A, B, \lambda)$

6: $\hat{u} = u_{ff} - kz = \ddot{q}_q - k(x - x_d) = \ddot{q}_d - k\left(\begin{bmatrix} q \\ \dot{q} \end{bmatrix} - \begin{bmatrix} q_d \\ \dot{q}_d \end{bmatrix}\right)$

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